Some standard results

If P(x) = $R(x)$ – $C(x)$ be the profit function, then

(i) Average profit:

$$AP = \frac{Total\ Profit}{Output} = \frac{P(x)}{x}$$

(ii) Marginal profit:

$$MP = \frac{dP}{dx} = \frac{d}{dx}(P(x)) = P'(x)$$

(iii) Marginal average profit:

$$MAP = \frac{d}{dx}(AP) = AP'(x)$$

(iv) Profit [$P(x)$] is maximum when $MR = MC$

6.1.2 Elasticity

Elasticity 'η' of the function $y = f(x)$ at a point x is defined as the limiting case of ratio of the relative change in y to the relative change in x.

$$\therefore \quad \eta = \frac{E_y}{E_x} = \lim_{\Delta x \to 0} \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\frac{dy}{y}}{\frac{dx}{x}}$$

$$\Rightarrow \quad \eta = \frac{x}{y} \cdot \frac{dy}{dx}$$

DO YOU NOW?

$$\eta = \frac{\frac{dy}{dx}}{\frac{y}{x}} = \frac{\text{Marginal quantity of } y \text{ with respect to } x}{\text{Average quantity of } y \text{ with respect to } x}$$

(i) Price elasticity of demand

Price elasticity of demand is the degree of responsiveness of quantity demanded to a change in price.

If x is demand and p is unit price of the demand function $x = f(p)$, then the elasticity of demand with respect to the price is defined as $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$.

(ii) Price elasticity of supply :

Price elasticity of supply is the degree of responsiveness of quantity supplied to a change in price.

If x is supply and p is unit price of the supply function $x = g(p)$, then the elasticity of supply with respect to the price is defined as $\eta_s = \frac{p}{x} \cdot \frac{dx}{dp}$.

NOTE

"price elasticity" is shortly called as "elasticity".

Some important results on price elasticity

(i) If $|\eta| > 1$, then the quantity demand or supply is said to be elastic.

(ii) If $|\eta| = 1$, then the quantity demand or supply is said to be unit elastic.

(iii) If $|\eta| < 1$, then the quantity demand or supply is said to be inelastic.

Remarks:

(i) Elastic : A quantity demand or supply is elastic when its quantity responds greatly to changes in its price. **Example: Cosumption of onion and its price.**

(ii) Inelastic: A quantity demand or supply is inelastic when its quantity responds very little to changes in its price. **Example: Consumption of rice and its price.**

(iii) Unit elastic : A quantity demand or supply is unit elastic when its quantity responds as the same ratio as changes in its price.

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